solution

Quiz 1 **03-04-2026**

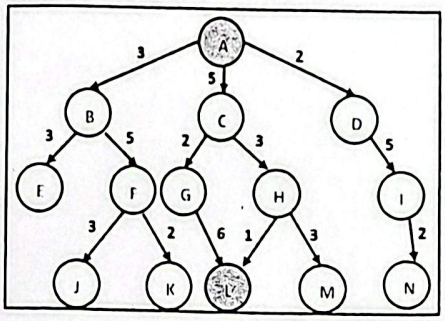
Artificial Intelligence **Mapping CLO2**

Student ID: **Total Marks: 20** **Obtained Marks:**

Student Name: **Section: E2** **Time: 25 mins**

Question 1:

Perform DFS on following graph, tell the path to reach goal. (Initial node is A and Goal node is L). **[marks 10]**



Step	Frontier	Expanded	Explored
	A	A	{}
	B, C, D	B	A
	E, F, C, D	E	A, B
	F, C, D	F	A B, E
	I, K, C, D	I	A B, E F
	K, C, D	K	A, B, E, F, I
	C, D	C	A, B, E, F, I, K
	G, H, D	G	A, B, E, F, I, K, C
	L, H, D	L	A, B E, F, I, K, C, G

A → C → G → L

Quiz 1 — 03-04-2026

Artificial Intelligence — Mapping CLO2

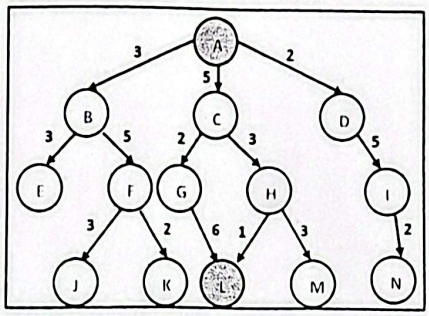
Student ID: — Total Marks: 20 — Obtained Marks:

Student Name: — Section: E2 — Time: 25 mins

Solution.

Question 1:

Perform **UCS** on following graph, tell the path to reach goal. (initial node is A and Goal node is L) **[marks 10]**



Step	Frontier	Expanded	Explored
	A	A	{ }
	~~A~~ (B,3) (A-C,5) (A-D,2)	D	A
	(A-B,3) (A-C,5) (A-D-I,7)	B	A,D
	(A-B-E,6) (A-B-F,8) ~~(A-C-G)~~	C	A,D,B
	~~(ADIE,7) (ABE,6) (ABF,8) (ACG,7) (ACH,8)~~	E	ADBC
	~~(ADIE,7) (ABE,8) (ACG,7) (ACH,8)~~	G	ADBCE
	~~(ADIE,7) (ABF,8) (ACH,8) (ACGL,13)~~	I	ADBCEG
	~~(ADIN,9) (ABF,8) (ACH,8) (ACGL,13)~~	F	I
	~~(ADIN,9) (ABFJ,11) (ABFK,10) (ACH,8) (ACGL,13)~~	H	F
	~~(ADIN,9) (ABFJ,11) (ABFK,10) (ACHL,9) (ACHM,11) (ACGL,13)~~	L	H

ADIN,9) (ABFJ,11) (ABFK,10) (ACHL,9) (ACHM,11) (ACGL,13)

ACHL — 9.